hozi

National Foresight & Decision Infrastructure
— first module: **Food Security**

"Fill the granary before the drought."

Nash Barara — Product & Design
Chipo — Strategy & Copy

POTRAZ AI4I 2026
DEVELOPMENT TRACK
ENVIRONMENT, CLIMATE & SUSTAINABILITY

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THE PROBLEM

The warning arrives too late. From outside.

- Drought and hunger are **rarely a surprise** — the signs build for weeks. The report arrives after the season is lost.
- Warnings come from **global agencies** — country-level, published outside the window for action.
- Even when a district is flagged, planners have **no shared way** to see where limited resources do the most good.

19 of 20 districts projected into high food-security risk by September — without early action.

7× return on timely, well-targeted anticipatory action (WFP).

The hard part is not knowing the drought is coming.
The hard part is **acting early enough, in the right places.**

WHAT HOZI DOES

From early warning to a ready plan.

- **District risk map** — ranked watch-list, updated automatically. See where the season is turning.
- **Time-slider forecast** — watch risk spread district by district, July through September.
- **Drill-down** — tap any district: rainfall, drying crops, trajectory. No black box.
- **Response Planner** — enter your available support; Hozi shows where it saves the most people, driest and least-served first.

In English, chiShona, isiNdebele.

WhatsApp and SMS alerts delivered to district officers on **2G** — no app install required.

THE PLATFORM PLAY

One engine. Many decisions.

Signals rainfall · crops · markets · satellite → **Risk Score** transparent · district-level · 0–100 → **Forecast** 1–3 months ahead · honest uncertainty band → **Ranked Response** where each dollar saves the most → **Playbooks** en · sn · nd · WhatsApp-ready

The pattern is always the same — build it once, apply it to every national risk:

Food Security	Cholera / Disease	Flood Displacement	Water Stress	Markets & Prices
Proven. Working now.	Next module — same engine	Next module — same engine	Roadmap	Roadmap

> No global tool helps Zimbabwe's own planners decide **where to act first**. Hozi does — in local languages, on a 2G signal, owned by Zimbabweans.

HOW AI IS USED

We keep AI out of the math. We put it into the language.

LAYER 1 — NUMBERS

Deterministic engine

Transparent, auditable, weighted risk model. Trained on **232 real IPC × CHIRPS observations** (58 districts, 2019–2020). Leave-one-district-out validation.

Every score is traceable arithmetic. No black box. No GPU required.

OLS · r = 0.484 · MAE 8.5pp vs 10.2pp global mean

LAYER 2 — REASONING

Claude API (read-only)

Receives only pre-computed numbers from the engine. Writes plain-language early-warning briefs in English, chiShona & isiNdebele. Answers planners' natural-language questions.

Parses incoming field officer reports — turns messy text into clean signals.

Advisory only · never generates data · never touches the model

> **The AI never touches the data. The numbers layer is pure deterministic math — so a Minister can audit every score.**

VALIDATION — WE PUBLISH OUR BASELINES

Ask other teams for theirs.

Leave-one-district-out validation · 232 real IPC × CHIRPS observations · 58 districts · 2019–2020

METHOD	R (PEARSON)	MAE (PP)	NOTES
Global-mean baseline	-0.557	10.2	Naive: always predict the panel mean
Persistence baseline	0.129	11.0	Predict last season's outcome
Rainfall-rule baseline	-0.179	10.7	Threshold on CHIRPS rainfall only
OLS model (Hozi)	**0.484**	**8.5**	**Beats all baselines · vegetation + market features next**

r = 0.81 is our *internal consistency check* (synthetic data). The table above is our **real validation** — on measured IPC outcomes the model never saw during training. We say exactly what each number is, and what it is not.

TECHNICAL ARCHITECTURE

Four layers. Every number traceable.

SOURCES

- **ZimVAC / IPC** — Phase outcomes
- **CHIRPS / TAMSAT** — Rainfall dekads
- **Agritex / Min. Agriculture** — Crops · pests · inputs
- **NASA / Copernicus** — NDVI satellite
- **GMB / ZMX** — Market prices

LAYER 0

n8n Automation (planned for pilot)
Scheduled pulls → engine trigger → threshold alert → WhatsApp dispatch to district officers

LAYER 1

Python engine (shipped)
Transparent risk model · Response Planner · OLS forecaster · LODO validation · baselines

14 unit tests pass · zero third-party Python deps

LAYER 2

Claude API (shipped)
Read-only · advisory briefs en/sn/nd · no data generation

LAYER 3

Static web app (shipped)
Vanilla HTML/CSS/JS · no build step · GitHub Pages · district map · time-slider · drill-down · Response Planner

> Zero third-party Python dependencies · 14 tests · every number traceable · no secrets in repo · MIT licensed

BUSINESS MODEL

Lean. Open-core.
Built to last without a big team.

Pilot monthly operating cost — 20 districts

Item	US$/mo
VPS — n8n + Python engine (2 GB)	20–40
GitHub Pages hosting	0
Claude API (playbook generation)	30–80
WhatsApp/SMS alert delivery	10–30
Domain + automated backups	5
Total	**65–155**

- **Phase 1 (0–12 mo):** Donor-funded pilot. Development partner covers operating costs; government operates and owns the data.
- **Phase 2 (12+ mo):** Per-domain institutional licensing. Each new module (health, climate/disaster) is a new institutional customer — same infrastructure.
- **Long-term:** Government line-item + ZCHPC national hosting. Operating cost moves off grants onto the National AI Strategy budget.

> Natural pilot funders: WFP Zimbabwe, FAO country office, UNDP. No partnership is claimed — these are the logical candidates.

DEPLOYMENT PLAN

Matabeleland South first. Then everywhere.

Pilot cluster

- **Districts:** Beitbridge · Gwanda · Mangwe (Matabeleland South)
- **Rationale:** Consistently highest food-insecurity risk in ZimVAC data · active ZimVAC and Agritex field presence · isiNdebele alignment
- **Users:** 10–20 district officers and ZimVAC/Agritex staff
- **Delivery:** WhatsApp/SMS alerts on 2G · no app install · static web app
- **Automation:** n8n scheduled CHIRPS pull → engine run → alert dispatch

Milestones

Day 30: Pilot MOU signed (or formal expression of interest) · demo URL live · compliance complete

Day 60: Live pilot running · alerts delivered · officer feedback logged

Day 90: Warning lead-time measured · adoption case study drafted · Fork A real-label model published · government handover runbook ready

n8n automation keeps the pipeline running with minimal human intervention. Clone repo → install → run. Full restore under 2 hours.

TEAM & THE ASK

Two Zimbabweans. AI-assisted. Fully disclosed.

Nash Barara

PRODUCT & DESIGN LEAD

11-year advertising career · Blender, Adobe Suite, AI creative tools · product vision, UX, pitch, deployment design · Curious Inq founder

Chipo

STRATEGY & COPY LEAD

Strategic framing · stakeholder narrative · policy alignment · ZimVAC & sector context · language copy (en/sn/nd)

WHAT WE ARE ASKING FOR

→ Incubation support and structured mentorship

→ Introduction to pilot MOU with ZimVAC or Ministry of Agriculture

→ Technical mentorship — live data feed integration and ZCHPC hosting pathway

AI build disclosure: Claude and Gemini were used throughout — engine validation, brief generation, document drafting. The architecture decisions, domain framing, and validation methodology are human-owned. We consider this the honest model for AI-assisted teams in 2026.

Prediction is a crowded field.
Prescription is not.

Demo · [Live URL — to be published on GitHub Pages]

Repo · github.com/[handle]/hozi

Proposal PDF · HOZI-TBD_AI4I_Proposal_Development.pdf (see /proposal/ai4i-2026/)

Tatenda. Siyabonga. Thank you.